
MODULE 1

COURSE OUTCOME:

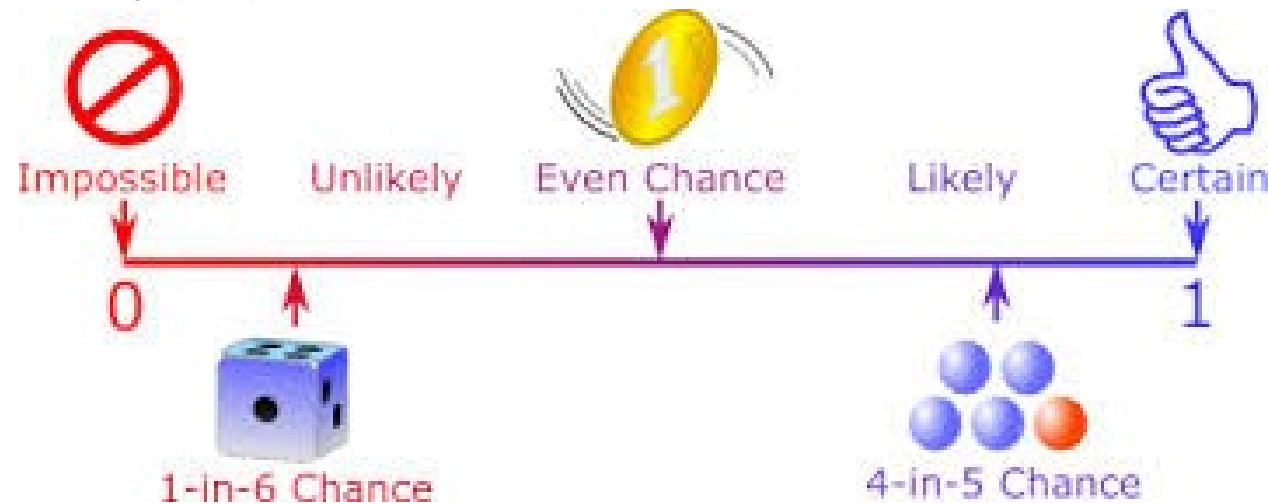
TO GAIN FUNDAMENTAL UNDERSTANDING OF PROBABILITY, CONDITIONAL PROBABILITY AND BAYES THEOREM.

MODULE CONTENTS:

- ❖ Probability Theory: Sample space, Events, Different approaches to probability
- ❖ Addition and multiplication theorems on probability
- ❖ Independent events, Conditional probability, Bayes Theorem

PROBABILITY

- **Probability** is the branch of mathematics concerning numerical descriptions of how likely an event is to occur, or how likely it is that a proposition is true.
- The probability of an event is a number between 0 and 1,
 - 0 indicates impossibility of the event
 - 1 indicates certainty (sure)



RANDOM EXPERIMENT, SAMPLE SPACE & EVENTS

- A **random experiment** is an **experiment** or a process (which can be repeated under same conditions any number of times) for which the outcome cannot be predicted with certainty.
 - Tossing a coin
 - Picking a card
- **Event** is one (or more) outcomes of a random experiment.
 - "The 5 of Hearts" (1 element)
 - "A King" (4 elements)
 - "A Face card" (12 elements)
 - "A Spade" (13 elements)
- A **sample space (S)** is a collection or a set of possible outcomes of a random experiment.



SAMPLE SPACE

A **sample space** is the set of all possible outcomes in an experiment.

Example:

Two coins are tossed. Represent the sample space for this experiment by making a list, a table, and a tree diagram.

(H – Head, T – Tail)

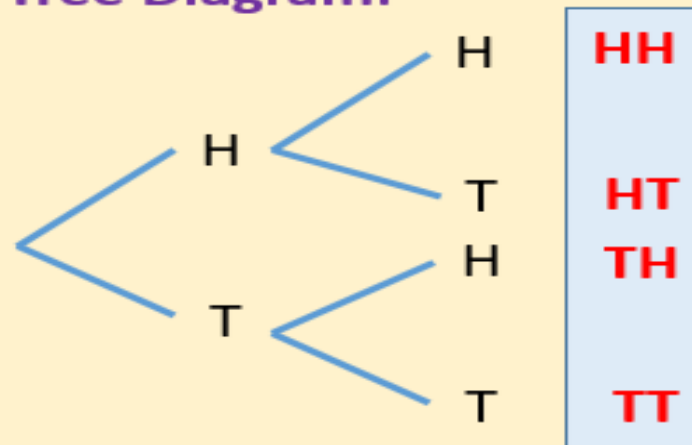
List:

HH HT TH TT

Table:

	H	T
H	HH	HT
T	TH	TT

Tree Diagram:



DIFFERENT TYPES OF SAMPLE SPACE

- A sample space S can be finite, countably infinite, uncountable infinite.
- A sample space S is either discrete or continuous.
- A discrete sample space S is either finite or countably infinite.
- **Finite** S , e.g.,
 - $S = \{1, 2, 3, 4, \cos(\pi/6), \sin(\pi/3)\}$
 - $S = \{T T, T H, H T, H H\}$
- **Countably infinite** S , e.g.,
 - $S = \{0, 1, 2, 3, \dots\}$
 - $S = \{T, H T, H H T, H H H T, \dots\}$.
- A continuous sample space S is **Uncountable infinite**
 - e.g., $S = \{\theta : 0 \leq \theta \leq 2\pi\}$

TYPES OF EVENTS

- **Null Event:**
($\emptyset$) A null event is an empty set, and has no outcomes.
- **Simple event** : Any event consisting of a single point of the sample space.
 - Ex: $S=\{1,2,3,4,5,6\}$ $E=\{5\}$
- **Compound Events**: If any event consists of more than one single point of the sample space
 - Ex: $S=\{1,2,3,4,5,6\}$ $E=\{2,3,5\}$

TYPES OF EVENTS

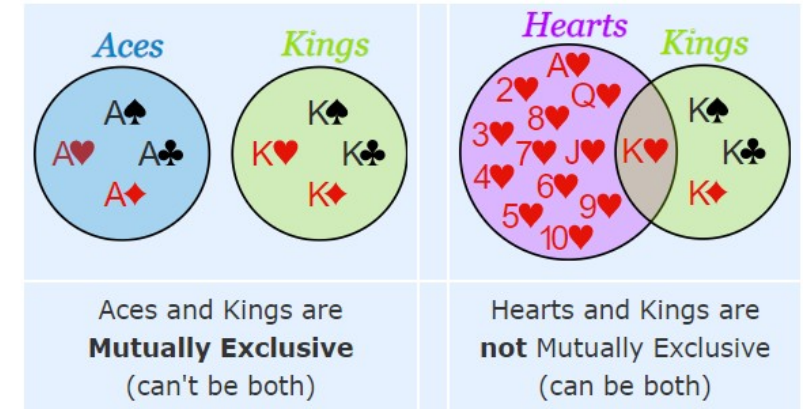
- **Independent Events and Dependent Events:**
 - If the occurrence of any event is completely unaffected by the occurrence of any other event, such events are known as an **independent event**
 - Ex: Tossing a coin 3 times to get a head in all three tosses
 - The events which are affected by other events are known as **dependent events**.
 - Ex: Drawing a King from a pack of cards 3 times
- **Replacement:** When we put each card **back** after drawing it the chances don't change, as the events are **independent**.
- **Without Replacement:** The chances will change, and the events are **dependent**.

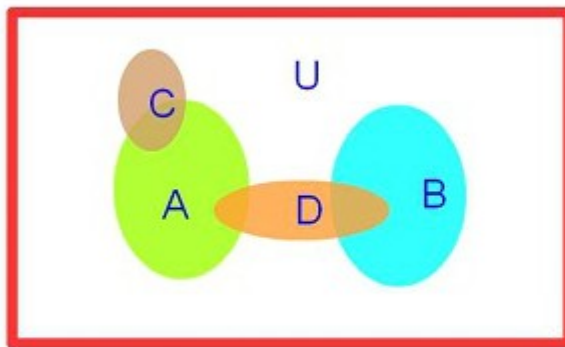
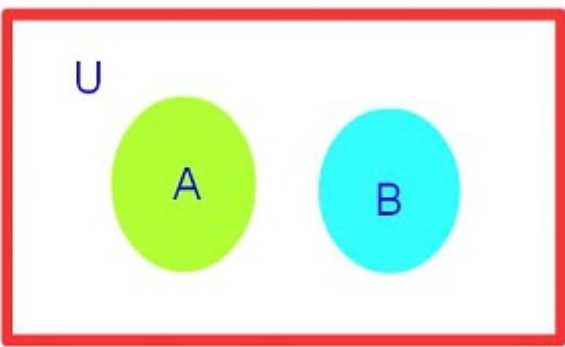
TYPES OF EVENTS

- **Complementary Events**
 - For any event E_1 there exists another event E_1' or E_1^c which represents the remaining elements of the sample space S .
 - Ex: $S=\{1,2,3,4,5,6\}$ $E_1=\{1,2,3\}$ $E_1^c=\{4,5,6\}$
- **Equally likely Events**: The events which are random and have an equal likelihood of occurrence.
 - Ex: During a coin toss you are equally likely to get heads or tails.
- **Favourable cases**: The cases that ensure the occurrence of an event.
- Ex: When a dice is thrown the favourable cases of getting an even number are $\{2,4,6\}$

MUTUALLY EXCLUSIVE & EXHAUSTIVE EVENTS

- **Mutually Exclusive** means we can't get both events at the same time.
- **Exhaustive events**: When a sample space is distributed down into some mutually exclusive events such that their union forms the sample space itself, then such events are called exhaustive events.
- When two or more events form the sample space collectively than it is known as collectively exhaustive events.
 - $E1=\{H\}$ $E2=\{T\}$
- Example:
 - Ex: $S=\{1,2,3,4,5,6,7,8,9,10\}$
 - $E1=\{1,3,5,7,9\}$
 - $E2=\{2,4,6,8\}$



Events A and B are mutually exclusive	Events A and B are mutually exclusive and exhaustive
	
The sample space includes outcomes of events other than A and B. The events A and B do not share any common outcome.	The sample space does not contain outcomes other than those belonging to A and B. The events A and B do not share common outcomes.

SAMPLE SPACE EXERCISE

1. We toss a coin until we see two consecutive tails. We record the total number of coin tosses.
2. A bag contains 44 balls: one is red, one is blue, one is white, and one is green. We choose two distinct balls and record their color in order.
3. A customer arrives at a bank and waits in the line. We observe TT , which is the total time (in hours) that the customer waits in the line. The bank has a strict policy that no customer waits more than 20 minutes under any circumstances.
4. Construct a sample space for the experiment that consists of rolling a single die. Find the events that correspond to the phrases “an even number is rolled” and “a number greater than two is rolled.”
5. Construct a sample space that describes all three-child families according to the genders of the children with respect to birth order.

MEASURES OF PROBABILITY

CLASSICAL APPROACH
RELATIVE FREQUENCY THEORY OF PROBABILITY
SUBJECTIVE PROBABILITY
AXIOMATIC APPROACH OF PROBABILITY

CLASSICAL APPROACH (PRIOR PROBABILITY)

- Classical Approach (Prior Probability): When all the outcomes are equally likely,

$$P(A) = \frac{\text{Number of outcomes favorable to } A}{\text{Number of possible outcomes}} = \frac{n}{N}$$

- What is the chance of getting a king in a draw from a pack of 52 cards?
 - $P(\text{getting a king}) = 4/52 = 1/13$
- Two coins are tossed simultaneously. What is the probability of getting a head and a tail?
- $S = \{HH, HT, TH, TT\}$
- Fav outcomes = $\{HT, TH\}$
- $P(\text{head and tail}) = 2/4 = 1/2$

RELATIVE FREQUENCY THEORY OF PROBABILITY

Relative frequency is the same as experimental probability.
We use relative frequency to predict probabilities from experimental data.

The experiment

This spinner was spun 40 times and the results recorded in this table:



Colour	Frequency
Blue	20
Yellow	10
Red	5
Green	5

Relative frequency

$$\frac{\text{frequency of event}}{\text{total number of trials}}$$

Event means one possible outcome; here, one colour on the spinner.

There were 20 blues recorded...

$$P(\text{blue}) = \frac{20}{40}$$

...out of 40 spins.

$$\text{Simplify: } P(\text{blue}) = \frac{20}{40} = \frac{2}{4} = \frac{1}{2}$$

SUBJECTIVE PROBABILITY

- Subjective probability is a type of probability derived from an individual's personal judgment or own experience about whether a specific outcome is likely to occur.
- It contains no formal calculations and only reflects the subject's opinions and past experience.
- Ex: Prediction of a sport event
- A doctor predicting 50% chance of need of a heart transplant based on his past experience

AXIOMATIC APPROACH OF PROBABILITY

Let, S be the sample space of any random experiment and let P be the probability of occurrence of any event.

1. For any event E , $P(E) \geq 0$
2. $P(S) = 1$
3. In case E and F are mutually exclusive events, then following equation will be valid:
$$P(E \cup F) = P(E) + P(F)$$

Two dice are thrown. Find the probability that :

- (a) the total of the numbers on the dice is 8,
- (b) the first die shows 6,
- (c) the total of the numbers on the dice is greater than 8,
- (d) the total of the numbers on the dice is 13,
- (e) both the dice show the same number,
- (f) the sum of the numbers shown by the dice is less than 5,
- (g) the sum of the numbers shown by the dice is exactly 6.

A card is taken at random from a full pack of 52 playing cards. What is the probability that it is:

- (a) a red card,
- (b) a 'Queen',
- (c) a red 'Ace',
- (d) the 'Seven of Hearts',
- (e) an even number?

A packet of sweets contains 18 *red* sweets, 12 *green* sweets and 10 *yellow* sweets. A sweet is taken at random from the packet. What is the probability that the sweet is:

- (a) *red*,
- (b) *not green*,
- (c) *green or yellow* ?

ADDITION THEOREM

Not Mutually Exclusive Events:

- if **A** and **B** are any two events then the probability of happening of at least one of the events is defined as $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

The probability that a student passes a Physics test is $(2/3)$ and the probability that he passes both Physics and English test is $(14/45)$. The probability that he passes at least one test is $(4/5)$. What is the probability that the student passes the English test?

ADDITION THEOREM

- **Mutually Exclusive Events:**
- If **A** and **B** are any two mutually exclusive events then $P(A \cap B) = 0$
 - $P(A \cup B) = P(A) + P(B)$.
- A bag contains 4 white, 3 black and 5 red balls. What is the probability of getting a white or a red ball at random in a single draw ?

CONDITIONAL PROBABILITY

- If A and B are two independent events the conditional probability of A given B is defined as
 - $P(A/B) = P(A \cap B)/P(B)$
- Similarly, conditional probability of B given A is defined as
 - $P(B/A) = P(A \cap B)/P(A)$

MULTIPLICATION THEOREM

- Multiplication Theorem states that if A and B are two **dependent events**, then the probability of their simultaneous occurrence is equal to the probability of one event multiplied by the conditional probability of the other.

$$P(A/B) = P(A \cap B)/P(B) \Rightarrow P(A \cap B) = P(B) * P(A/B)$$

$$P(B/A) = P(A \cap B)/P(A) \Rightarrow P(A \cap B) = P(A) * P(B/A)$$

MULTIPLICATION THEOREM

- If two events A and B are **independent**, then the probability that both will occur is equal to the product of the respective probabilities.

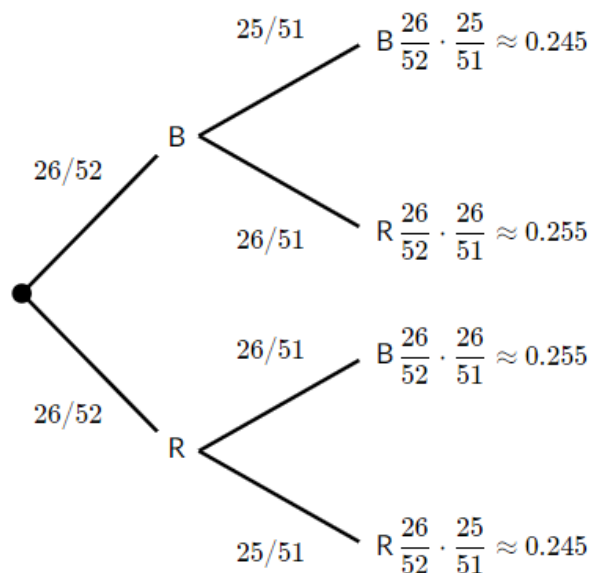
$$P(A \cap B) = P(A) * P(B)$$

- The probability that X and Y will be alive ten years hence is 0.5 and 0.8 respectively. What is the probability that both of them
- i) will be alive ten years hence ?
- ii) will not be alive ten years hence ?
- Also find the probability that either of them will be alive after 10 years.

MULTIPLICATION THEOREM

- Consider drawing two cards, without replacement, from a standard deck of 52 cards. That means we are drawing the first card, leaving it out, and then drawing the second card. What is the probability that both cards selected are black?
- $P(\text{both black card}) = P(\text{first black card}) \cdot P(\text{second black card} / \text{first card is black})$

$$= 26/52 \cdot 25/51 = \frac{1}{2} \cdot \frac{25}{51} = \frac{25}{102} \approx 0.245$$



MULTIPLICATION THEOREM

- A university has to select an examiner from a list of 50 persons, 20 of them women and 30 men, 10 of them knowing Hindi and 40 not. 15 of them being teachers and the remaining 35 not. What is the probability of the University selecting a Hindi-knowing women teacher ?
- Probability of selecting a women $= 20/50 = 2/5$
- Probability of selecting a teacher $= 15/50 = 3/10$
- Probability of selecting a Hindi-knowing candidate $= 10/50 = 1/5$
- Probability (selecting a hindi knowing women teacher) $= 2/5 * 3/10 * 1/5 = 3/125$

A problem in statistics is given to three students A, B, C whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$ respectively. What is the probability that the problem will be solved ?

The probability that A speaks truth is $\frac{4}{5}$, while its probability for B is $\frac{3}{4}$, the probability that they contradict each other when asked to speak on a fact is:

ODDS

- Odds relate the chances in favour of an event to the chances against it.
- **If the odds that an event A occurs is $a:b$, then**

$$P(A) = \frac{a}{a+b}$$

Example: If the odds of James winning or losing the Game are 9:2, what is the subjective probability that he will win?

$$P(\text{James winning}) = 9/11$$

If two dice are thrown, what is the probability that the sum is

- (a) greater than 9, and
- (b) neither 7 nor 11?

A card is drawn from a well-shuffled pack of playing cards. What is the probability that it is either a spade or an ace?

The Odds against manager X settling the wage dispute with the workers are 8:6 and odds in favor of manager Y settling the same dispute are 14:16.

- (a) what is the chance that neither settles the dispute, if they both try, independently of each other?
- (b) what is the probability that the dispute will be settled?

BAYES' THEOREM

INVERSE PROBABILITY

Bayes' Theorem is a fundamental result in probability theory that describes the probability of an event based on prior knowledge of conditions that might be related to the event.

BAYES THEOREM

Formula For Bayes' Theorem

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B|A)}{P(B)}$$

where:

$P(A)$ = The probability of A occurring

$P(B)$ = The probability of B occurring

$P(A|B)$ = The probability of A given B

$P(B|A)$ = The probability of B given A

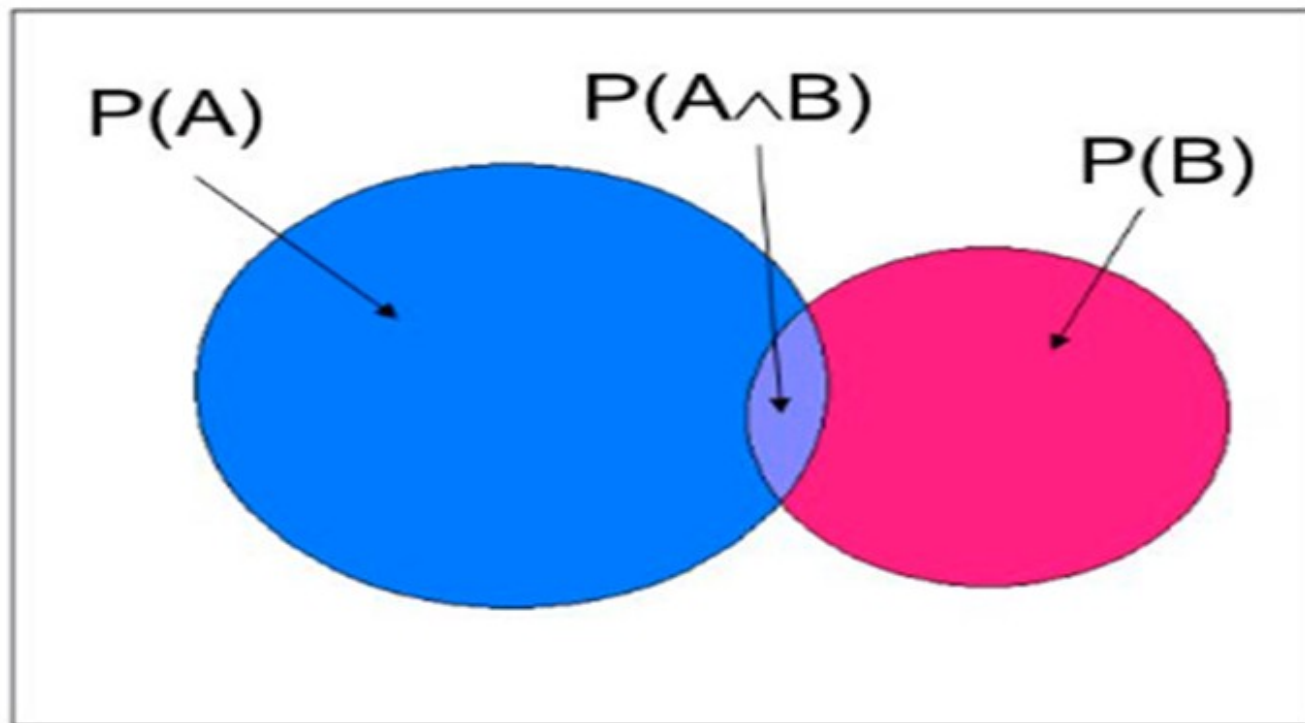
$P(A \cap B)$ = The probability of both A and B occurring

$$\text{posterior} = \frac{\text{prior} \times \text{likelihood}}{\text{evidence}}$$

- Bayes' theorem allows you to update predicted probabilities of an event by incorporating new evidence

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A')P(A')}$$

BAYE'S THEOREM



$$P(A|B) = P(A \cap B) / P(B)$$

$$P(B|A) = P(A \cap B) / P(A)$$

$$P(A|B) = P(B|A)P(A) / P(B)$$

(Bayes formula for two events)

There are two identical urns containing respectively 6 black and 4 red balls, 2 black and 2 red balls. An urn is chosen at random and a ball is drawn from it.

(I) find the probability that the ball is black

(ii) if the ball is black, what is the probability that it is from the first urn?

- $P(\text{Black}) = ? \quad P(U1/B) = ?$
- $P(B/U1) = 6/10$
- $P(B/U2) = 2/4 = 1/2$
- $P(\text{Black}) = P(U1).P(B/U1) + P(U2)P(B/U2)$
 $= \frac{1}{2} * 6/10 + \frac{1}{2} * 2/4$
 $= 3/10 + 1/4 = 11/20$

$$P(U1/B) = \frac{P(B/U1)*P(U1)}{P(B)} = \frac{6/10 * 1/2}{11/20} = \frac{6/20}{11/20} = 6/11$$

You randomly choose a treasure chest to open, and then randomly choose a coin from the treasure chest. If the coin you choose is gold, what is the probability you chose chest A?

100 gold coins



A

50 gold + 50 silver coins



B

A company has two plants to manufacture scooters. Plant I manufactures 80% of the scooters and plant II manufactures 20%. At plant I, 85 out of 100 scooters are rated as of standard quality and at plant II, 65 out of 100 scooters are rated as of standard quality. A scooter is chosen at random and is found to be of standard quality. What is the probability that it has come from plant II?

BAYE'S THEOREM

Bayes' theorem.. Let $A_1, A_2, \dots, A_n$ be a set of mutually exclusive events that together form the sample space S . Let B be any event from the same sample space, such that $P(B) > 0$. Then,

$$P(A_k | B) = \frac{P(A_k \cap B)}{[P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_n \cap B)]}$$

Note: Invoking the fact that $P(A_k \cap B) = P(A_k)P(B | A_k)$, Baye's theorem can also be expressed as

$$P(A_k | B) = \frac{P(A_k) P(B | A_k)}{[P(A_1) P(B | A_1) + P(A_2) P(B | A_2) + \dots + P(A_n) P(B | A_n)]}$$

A speaks truth 3 times out of 4 while B does so 7 times out of 10. A ball is drawn at random from a bag containing one black ball and five other balls of different colors. Both A and B report that a black ball has been drawn from the bag. What is the probability of their assertion being true?

In an objective type examination, each question has four options of which only one is correct. A student taking the examination either guesses or copies or knows the answer to any question. The probability that he guesses is $\frac{1}{3}$ and that the he copies is $\frac{1}{6}$. the probability that he answers correctly if he copies is $\frac{1}{8}$. If an answer is given to be correct, find the probability that the student knew the answer.

SUMMARY

Classical Probability	$P(E) = \frac{\text{Favourable}}{\text{Total Outcomes}}$
Complement	$P(A^c) = 1 - P(A)$
Addition Theorem	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Mutually Exclusive	$P(A \cup B) = P(A) + P(B)$
Conditional Probability	$P(A B) = \frac{P(A \cap B)}{P(B)}$
Multiplication (Dependent)	$P(A \cap B) = P(A)P(B A)$
Multiplication (Independent)	$P(A \cap B) = P(A)P(B)$
Bayes' Theorem	$P(A B) = \frac{P(B A)P(A)}{P(B)}$
Odds in Favour	$P = \frac{a}{a + b}$